

# LENNOX SAMA KEEBLE

## Curriculum Vitae

Last updated: August 4, 2026.

### Education

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#### Wake Forest University

*Doctor of Philosophy (Ph.D.) in Physics. Supervisor: Professor Alejandro Cárdenas-Avendaño*

**2026 - 2031**

*Winston-Salem, USA*

#### University of Cambridge

*Master of Advanced Study (MASt) in Theoretical Physics. Supervisor: Dr. Lorenzo Gavassino*

**2025 - 2026**

*Cambridge, UK*

#### Princeton University

*Bachelor of Arts (B.A.) in Physics. Supervisor: Professor Frans Pretorius*

**2020 - 2025**

*Princeton, USA*

### Honors, Scholarships & Awards

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- Cambridge Trust Master's Studentship, University of Cambridge, 2025.
- Academic Futures Graduate Scholarship, University of Oxford, 2025. (Declined.)
- Allen G. Shenstone Prize in Physics, Princeton University, Department of Physics, 2025.
- Manfred Pyka Memorial Prize, Princeton University, Department of Physics, 2022.

### Publication Summary

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- **5** (submitted/accepted) refereed papers; **4** as lead author
- **Citations:** > **90**; **h-index:** 5 ([Google Scholar](#))
- **Papers by research area:** relativistic hydrodynamics (**2**); black hole imaging (**2**); extreme mass-ratio inspirals (**1**)
- Links to academic metrics: [Google Scholar](#), [INSPIRE](#)

### Invited Talks, Conferences & Workshops

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bold **numbers** indicate a contributed talk, bold **text** indicates an invited talk

8. IGWN-PI School on Gravitational Waves, Perimeter Institute for Theoretical Physics, 2026.
7. Relativistic Astrophysics Summer School, Caltech, 2025.
6. Nuclear Physics of MultiMessenger Mergers (NP3M) Summer School, Indiana University, 2025.
5. **Los Alamos Astrophysics Distinguished Seminar Series, Center for Nonlinear Studies, Los Alamos National Laboratory, 2025.**
4. Black Hole Mimickers: From Theory to Observation, Princeton Center for Theoretical Science & Princeton Gravity Initiative, 2025.
3. CU<sup>2</sup>MIP Undergraduate Conference, College Park, USA, 2024.
2. American Physical Society April Meeting, 2024.
1. Taking it to the extreme: Symmetries and dynamics of extremal black holes, Princeton Center for Theoretical Science & Princeton Gravity Initiative, 2024.

### Professional Experience

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#### Investment Associate Intern

*Bridgewater Associates*

**Summer 2024**

*Westport, Connecticut, USA*

#### Summer Business Analyst

*McKinsey & Company*

**Summer 2022**

*San Francisco, California, USA*

## Miscellaneous

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**Programming Languages:** julia., Python, Mathematica & Slurm.

**Web Links:** [Webpage](#), [GitHub](#).

## List of Publications

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<sup>†</sup> indicates first-authored papers or alphabetical ordering

### Refereed (submitted/accepted)

- [5] *Radial oscillations of viscous stars.*  
<sup>†</sup> **L. S. Keeble** and J. Redondo-Yuste.  
[Phys. Rev. D 113, 124058 \(2026\)](#) [arXiv: 2603.23622 \(gr-qc\)](#).
- [4] *Estimating high-order time derivatives of Kerr orbital functionals.*  
<sup>†</sup> **L. S. Keeble** and A. Cárdenas-Avendaño.  
[Physical Review D 112, 084048 \(2025\)](#) [arXiv: 2508.08888 \(gr-qc\)](#).
- [3] *Inferring black hole spin from interferometric measurements of the first photon ring: A geometric approach.*  
<sup>†</sup> **L. S. Keeble**, A. Cárdenas-Avendaño, and D. C. M. Palumbo.  
[Physical Review D 111, 103042 \(2025\)](#) [arXiv: 2502.20312 \(astro-ph.HE\)](#).
- [2] *First-order viscous relativistic hydrodynamics on the two-sphere.*  
<sup>†</sup> **L. S. Keeble** and F. Pretorius.  
[Physical Review D 112, 124034 \(2025\)](#) [arXiv: 2508.20998 \(gr-qc\)](#).
- [1] *Assessing the impact of instrument noise and astrophysical fluctuations on measurements of the first black hole photon ring.*  
A. Cárdenas-Avendaño, **L. Keeble**, and A. Lupsasca.  
[Physical Review D 109, 124052 \(2024\)](#) [arXiv: 2404.01083 \(gr-qc\)](#).

### White Papers and Conference Proceedings

- [1] *The Black Hole Explorer: photon ring science, detection, and shape measurement.*  
A. Lupsasca, A. Cárdenas-Avendaño, D. C. M. Palumbo, M. D. Johnson, S. E. Gralla, D. P. Marrone, P. Galison, P. Tiede, and **L. Keeble**.  
[Proc. SPIE Int. Soc. Opt. Eng. 13092, 130926Q \(2024\)](#) [arXiv: 2406.09498 \(gr-qc\)](#).